CBSE – CLASS XII PRACTICE QUESTION PAPER

Subject: CHEMISTRY

Chapter: Alcohols, Phenols and Ethers

Time Allowed: 1 Hour Maximum Marks: 30

General Instructions:

1. All questions are compulsory.
2. The paper has 10 questions in three sections: Section A (Q1–4, 2 marks each), Section B (Q5–8, 3 marks each) and Section C (Q9–10, 5 marks each).
3. Show all steps in numerical problems with proper units. Draw neat, labelled diagrams wherever asked.

SECTION A (Q1–Q4) – 2 marks each

Q1. Write the IUPAC names of the following compounds: (i) $CH_3–CH(CH_3)–CH(OH)–CH_3$ (ii) $CH_3–O–CH_2–CH_2–CH_3$ (iii) $C_6H_5–O–CH_2CH_3$ (iv) $HOCH_2–CH(OH)–CH_2OH$ **[2]**

Q2. (a) Why do alcohols have much higher boiling points than ethers and hydrocarbons of comparable molecular masses? Draw the hydrogen bonding between alcohol molecules. (b) Arrange in increasing order of boiling point: ethanol, methoxymethane, propane. **[2]**

Q3. Write the chemical equations for: (i) hydroboration–oxidation of propene; (ii) reaction of phenol with bromine water. **[2]**

Q4. Phenol is more acidic than ethanol. Explain with the help of resonance structures of the phenoxide ion. **[2]**

SECTION B (Q5–Q8) – 3 marks each

Q5. Write the mechanism of acid-catalysed dehydration of ethanol to ethene (conc. H_2SO_4, 443 K). Why is the temperature kept at 443 K and not at 413 K? **[3]**

Q6. 4.7 g of phenol is treated with excess bromine water to give 2,4,6-tribromophenol. Calculate (i) the number of moles of phenol used, (ii) the mass of 2,4,6-tribromophenol formed, and (iii) the mass of bromine required, assuming 100% yield. (Atomic masses: C = 12, H = 1, O = 16, Br = 80) **[3]**

Q7. 23 g of ethanol on dehydration with conc. H_2SO_4 at 443 K gives ethene in 80% yield. Calculate the mass of ethene obtained and its volume at STP. (Atomic masses: C = 12, H = 1, O = 16; molar volume at STP = 22.4 L) **[3]**

Q8. Give one chemical test to distinguish between each of the following pairs: (i) phenol and ethanol (ii) propan-1-ol and propan-2-ol (iii) ethanol and methoxymethane. **[3]**

SECTION C (Q9–Q10) – 5 marks each

Q9. (a) How is phenol manufactured from cumene? Write the equations involved and name the by-product of commercial importance. (b) Why does the reaction of sodium ethoxide with tert-butyl bromide not give ethyl tert-butyl ether? Suggest a suitable combination of reactants for preparing $(CH_3)_3C–O–C_2H_5$ by Williamson synthesis. **[5]**

Q10. Write the chemical equations for: (a) Reimer–Tiemann reaction of phenol, (b) Kolbe's reaction, and (c) the reaction of anisole with HI. Give the reason for the products in (c). **[5]**